

Exploring Performance Landscapes in Text Classification: Effects of Feature Representation and Regularization

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Abstract

This project studies how logistic regression behaves across different feature representations and regularization settings in title-based fake news classification. Rather than focusing only on the single best tuned model, the analysis maps performance across broad grids of hyperparameters and interprets the resulting heatmaps as performance landscapes. Using article titles from a fake versus real news dataset, we compare Bag-of-Words and TF-IDF representations under unigram, unigram-plus-bigram, and bigram-only feature spaces, with both L1 and L2 regularization. Performance is evaluated using F1 score and AUC, and train-test F1 gap is used to study generalization behavior. The heatmaps reveal several consistent structures: L1 produces a sharp low- C collapse, L2 changes more smoothly, unigram and unigram-plus-bigram models behave similarly, and bigram-only models are both weaker and less stable. Taken together, the results suggest that the most useful region is not the most weakly regularized extreme, but a moderate- C plateau where predictive performance is already strong and train-test divergence remains smaller.

1 Introduction

In text classification, it is common to choose a representation, tune a few hyperparameters, and report the best-performing model. That approach is useful, but it often leaves an important question unanswered: how does the model behave across the parameter space, and not just at one selected setting? In practice, choices such as Bag-of-Words versus TF-IDF, unigram versus bigram features, or strong versus weak regularization are often made because they seem reasonable or because they work well in a limited search. What is usually missing is a clear picture of how the model behaves across the full range of those choices. A heatmap-based analysis makes it possible to study these choices more systematically.

Instead of viewing logistic regression as a model that simply needs to be tuned, we treat it as something whose behavior can be mapped. The goal is to examine how feature representation, regularization, and feature budget shape model behavior in a text classification setting. Instead of focusing only on the single best score, we study where performance is strong, where it collapses, where it stabilizes, and where it becomes less reliable.

The task is fake versus real news classification using article titles. We compare Bag-of-Words and TF-IDF representations under unigram, unigram-plus-bigram, and bigram-only feature spaces,

with both L1 and L2 regularization. Performance is evaluated using F1 score and AUC, and the train-test F1 gap is used to study generalization behavior. Looking at all three matters because they capture different aspects of model behavior. F1 evaluates the final classifications, AUC measures how well the model keeps the two classes apart overall, and the train-test gap helps identify settings that are strong on the training data but fragile once generalized beyond it.

The main question of this study is therefore not simply which logistic regression setting performs best. Instead, the goal is to understand how the model behaves across the parameter space, and what those patterns suggest about the interaction between representation, regularization, and the data.

2 Data Description and Preliminary Analysis

2.1 Dataset and Scope

The data come from a fake versus real news dataset available on Kaggle. The real articles were collected from Reuters, while the fake articles were gathered from sources identified as unreliable by Politifact and Wikipedia. The full dataset includes article titles, article text, subject labels, and publication dates, with most articles coming from 2016 to 2017 and focusing mainly on political and world news topics.

For this project, we use only article titles and work with a balanced subset of 10,000 observations. This keeps the analysis computationally manageable while also providing a cleaner setting for comparing feature representations and regularization choices. The final subset is exactly balanced between fake and real titles (5,000 fake and 5,000 real). This was done so that the later heatmap patterns would be easier to interpret and less likely to be driven by baseline class proportions.

2.2 Title Length

Because the analysis uses titles only, the text inputs are short relative to full news articles. This matters because it limits how much information can be captured by longer n-grams and helps explain why some representations are more sensitive than others.

Figure 1 shows the distribution of title lengths by class. This helps establish the scale of the inputs and give context for the later feature-representation results.

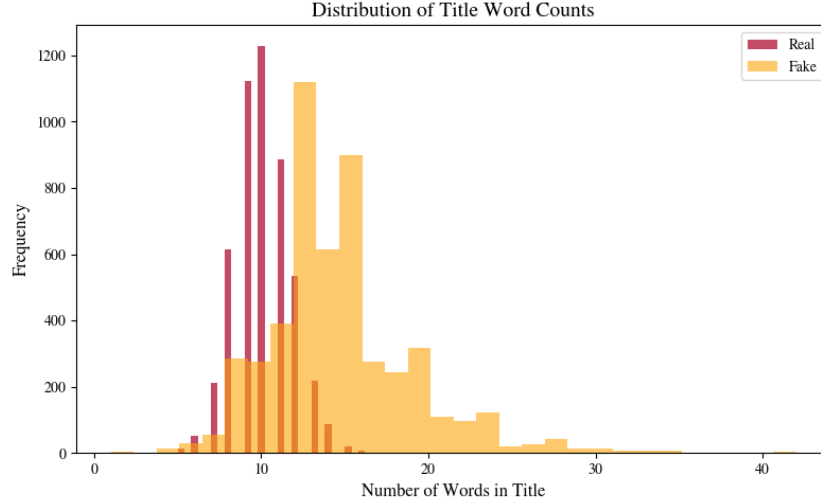


Figure 1: Distribution of title lengths by class.

2.3 Common Words

Looking at the most common words in each class suggests that fake and real titles differ in both vocabulary and style. Real titles use more institutional and geopolitical terms, while fake titles more often include attention-grabbing words and prominent political names. Since certain words lean strongly towards one class, this helps explain why logistic regression performs well in this setting and why unigram-based representations capture much of the useful signal.

Figure 2 shows the most common words in real and fake titles after light preprocessing.

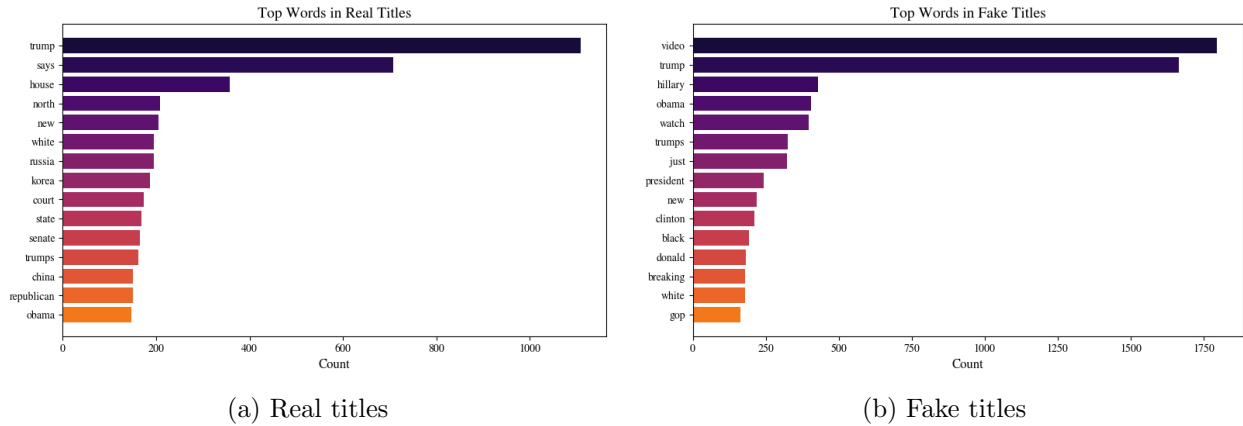


Figure 2: Most common words by class.

These exploratory figures provide the basic context needed to interpret the performance landscapes in the following sections.

3 Methods

3.1 Feature Representations

The article titles were converted into numerical features using two standard text representations: Bag-of-Words and TF-IDF. In a Bag-of-Words representation, each feature records how often a word or phrase appears. In a TF-IDF representation, terms that appear in many titles receive less weight, while terms that are more distinctive receive more weight.

For each representation, three n-gram settings were considered: unigram only, unigram plus bi-gram, and bigram only. These choices were included to separate the value of single-word information from the value of phrase-level information. In particular, comparing unigram and unigram-plus-bigram representations reveals whether bigrams add useful signal beyond what unigrams already provide, while bigram-only models isolate the effect of phrase-level features on their own.

3.2 Logistic Regression and Regularization

All models in the main analysis were fit using logistic regression. For an input feature vector x , the model estimates the probability that the title belongs to the fake-news class:

$$P(y = 1 \mid x) = \sigma(w^\top x + b),$$

where $\sigma(\cdot)$ is the logistic function, w is the vector of feature weights, and b is an intercept term. In this setting, each weight measures how strongly a word or phrase pushes the prediction toward one class or the other.

Regularization is used to keep the model from relying too heavily on any one set of features. This is especially important in text classification, where the number of possible features can be large relative to the amount of data. The regularization strength is controlled by the parameter C , where smaller values of C mean stronger regularization and larger values mean weaker regularization.

Two penalties were studied – L1 and L2. L1 regularization can force many coefficients to exactly zero, which produces a sparse fitted model. L2 regularization shrinks coefficients continuously but usually keeps more of them active. Since these penalties control model complexity in different ways, they can produce different patterns across the performance landscapes.

3.3 Hyperparameter Grid and Model Fitting

The heatmaps were built over a two-dimensional grid of hyperparameters:

- the regularization parameter C ,
- the maximum number of retained features, `max_features`.

The values of C were chosen on a log scale so that the analysis covered very strong and very weak regularization without concentrating too heavily on only one part of the range. The values

of `max_features` were chosen over a broad linear range to show how performance changed as the vocabulary size increased.

The balanced title-only subset was split into training and testing sets using an 80/20 stratified train-test split so that class balance was preserved in both sets. For each combination of representation, n-gram setting, penalty, C , and `max_features`, the vectorizer was fit on the training data only and then applied to the test data. Logistic regression was then fit on the training matrix and evaluated on the test matrix. Repeating this process across the full grid produced one heatmap for each representation and penalty combination.

3.4 Evaluation Metrics

The analysis uses three views of model behavior: Test F1, Test AUC, and the train–test F1 gap.

F1 score is used as the main measure of classification performance. It depends on the model’s final class predictions and balances precision and recall. AUC is used to measure how well the model keeps the two classes apart overall. Unlike F1, it does not depend on just one final cutoff for classification. The train–test F1 gap, defined as training F1 minus testing F1, is used to study stability and generalization. Larger gaps suggest that the model is fitting the training data more strongly than the test data.

These three views were used together because they do not measure the same thing. A pattern that appears in both F1 and AUC is more likely to reflect a real change in how well the model distinguishes the classes. A pattern that appears mainly in F1 may be more tied to the model’s final classification decisions. The gap heatmaps were then used to distinguish regions that perform well and generalize well from regions that perform well but are less stable.

3.5 Heatmap Interpretation

For each representation and penalty combination, separate heatmaps were constructed for Test F1, Test AUC, and the F1 gap over the $(C, \text{max_features})$ grid. The goal was not only to locate strong settings, but to study the overall shape of the parameter space.

Interpretation focused on broad, repeated structures such as collapse regions, transition zones, plateaus, and unstable regions. Small local differences were treated more cautiously, especially since panel-specific color scales can make small numerical changes look more dramatic than they are.

4 Results

The heatmaps show that model behavior is highly structured across the parameter space. In most settings, the dominant changes occur along the regularization axis rather than the feature-budget axis, which means that the choice of C often matters more than `max_features` once the vocabulary is no longer extremely small. At the same time, this broad geometry does not look the same for all representations: unigram-inclusive models tend to show wide high-performing regions,

while bigram-only models are lower-performing and more sensitive to the retained feature budget. The same large-scale transitions also appear in both F1 and AUC, which suggests that the main structures in the heatmaps are real features of the landscape rather than artifacts of a single metric. The subsections below examine these patterns in more detail.

4.1 L1 and L2 Create Different Performance Regimes

The clearest distinction in the heatmaps is between L1 and L2 regularization. Under L1, many of the F1 heatmaps contain a sharp left-side failure region at very small values of C . This appears as a nearly vertical cutoff, with performance collapsing across nearly the entire range of `max_features`. The corresponding AUC heatmaps show that the same region often falls to approximately 0.5, which means that the model is largely failing to keep the two classes apart. A plausible explanation is that, in this short and sparse title-based setting, strong L1 regularization removes so many useful lexical features that the model’s score collapses back toward the intercept. Once that happens, the classifier is no longer using enough active signal from the title to separate the two classes. Since the dataset is balanced, this pushes predicted probabilities toward the middle and creates a near-complete failure regime rather than a gradual decline.

L2 behaves differently. At small values of C , performance weakens, but it does not collapse in the same abrupt way. Instead, the transition from poor to strong performance is generally smoother. This means that the two penalties create different kinds of low- C behavior: L1 produces a sharper failure regime, while L2 produces a more gradual change across the same part of the parameter space.

This is an important structural difference between the penalties. It suggests that, in this setting, L1 creates a more abrupt threshold between unusable and usable regions, while L2 allows the model to remain partially functional over a broader range of regularization strengths. More detailed differences between the penalties, especially those involving metric-specific patterns, are discussed later.

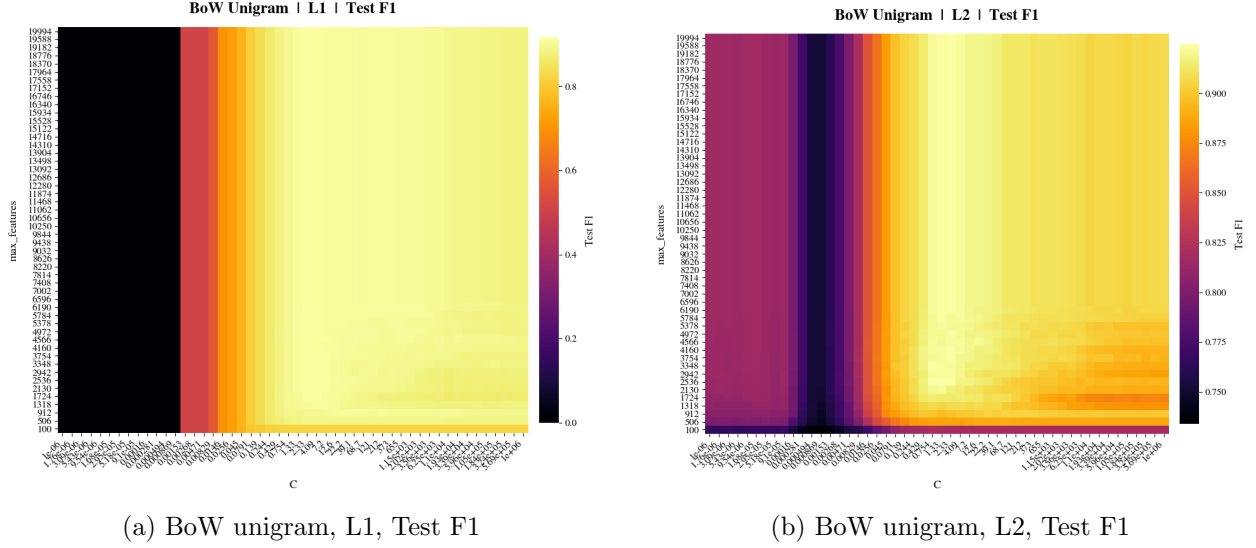


Figure 3: L1 and L2 produce different large-scale landscape shapes. L1 shows a sharp low- C collapse, while L2 changes more gradually.

4.2 Unigram and Unigram-Plus-Bigram Models Are Structurally Similar

A strong pattern across the heatmaps is that unigram and unigram-plus-bigram models behave very similarly. Their F1 and AUC surfaces share the same broad transitions, similar plateaus, and similar sensitivity to `max_features`. This suggests that, in this title-only setting, most of the useful signal is already captured by individual words. Adding bigrams changes the representation, but it does not substantially change the overall geometry of the performance landscape.

This result is consistent with the exploratory word-frequency plots. The titles appear to differ strongly in vocabulary, and many of the most common terms are already informative at the single-word level. Since titles are short, there may simply not be enough additional phrase-level information for unigram-plus-bigram models to separate themselves strongly from unigram-only models.

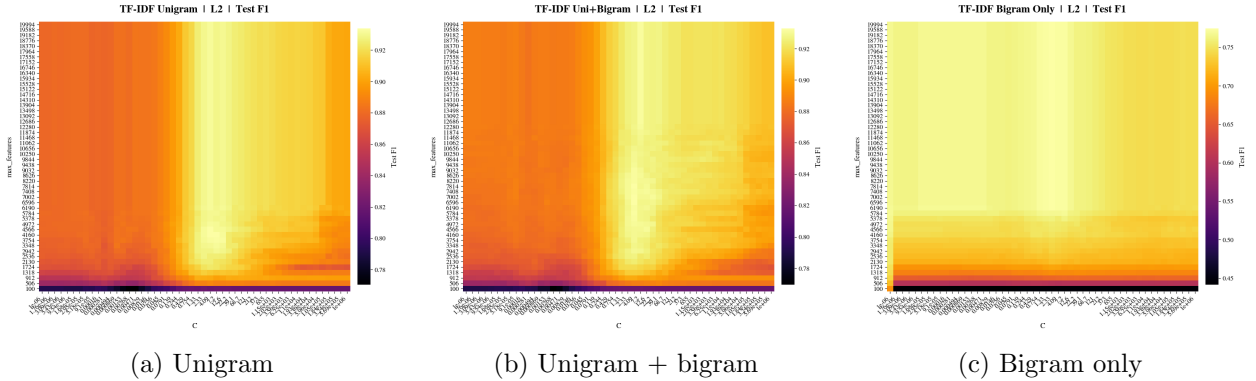


Figure 4: Under TF-IDF with L2 regularization, unigram and unigram-plus-bigram models have very similar F1 landscapes, while bigram-only behaves differently.

4.3 Bigram-Only Models Are More Sensitive and Less Stable

The bigram-only heatmaps show a different kind of behavior. Their overall performance is lower, but more importantly, they are much more sensitive to the size of the retained vocabulary. Unlike unigram-inclusive models, which level off once a moderate feature budget is reached, bigram-only models continue to depend more strongly on `max_features`. This suggests that phrase-level features alone do not provide a dense enough representation in this title-based setting.

The gap heatmaps strengthen this interpretation. Bigram-only models are not only weaker in F1 and AUC, but also more fragile across broad parts of the grid. In other words, the issue is not just that phrase-only representations contain less signal. They also produce a less stable model, especially once regularization weakens and the classifier begins fitting sparse phrase patterns more aggressively.

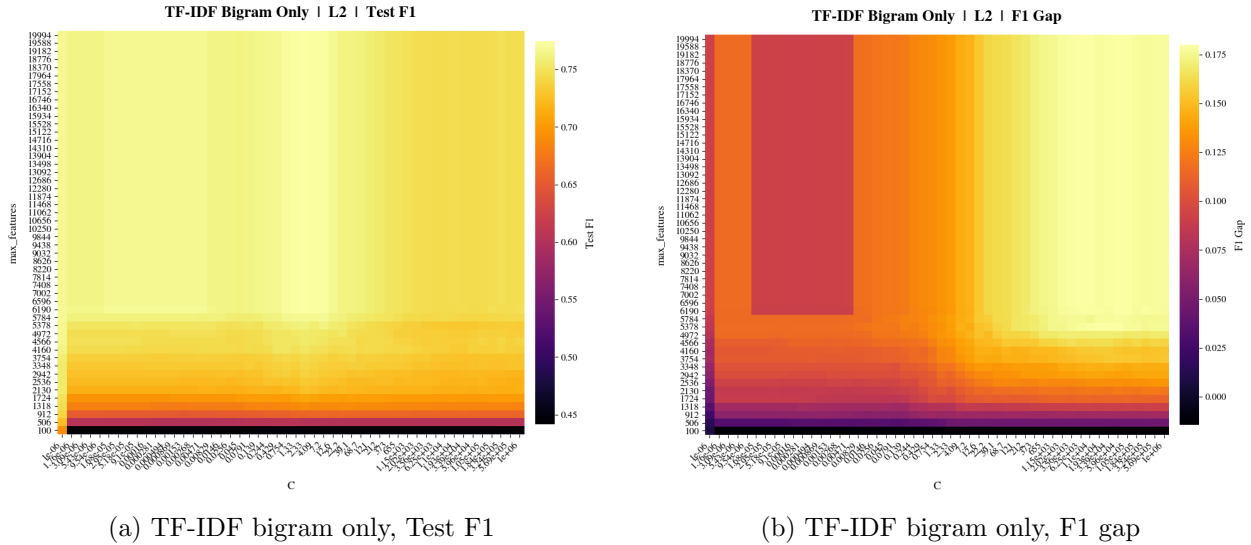


Figure 5: Bigram-only models are not only lower-performing overall, but also less stable across the parameter space.

4.4 Vocabulary Saturation and the Low-Feature, Weak-Regularization Corner

Across most of the heatmaps, the dominant variation occurs along the regularization axis, which means that C matters more than `max_features` once the vocabulary is no longer extremely small. For unigram and unigram-plus-bigram models, performance usually improves quickly as the feature budget increases from very small values, then levels off once a moderate vocabulary size is reached. This suggests that most of the useful lexical signal is already captured after a relatively modest number of features.

One recurring exception appears in the bottom-right corner of many heatmaps, where `max_features` is relatively restricted and C is very large. In this region, both F1 and AUC often drop while the train-test F1 gap increases. This indicates that, under weak regularization, a limited feature budget can become not only a limitation, but a liability. With too few retained features, the model is

forced to rely on a narrow set of lexical cues, and weak regularization allows those cues to be fit too aggressively to the training data. As a result, the model becomes both weaker and less stable in this corner of the parameter space.

This corner effect is useful because it shows that the role of `max_features` is not completely exhausted once the main heatmap saturates. Over most of the grid, the maps are largely vertical, but in the low-feature, weak-regularization corner, horizontal structure reappears. That makes this region a small but meaningful exception to the broader pattern that regularization dominates vocabulary size.

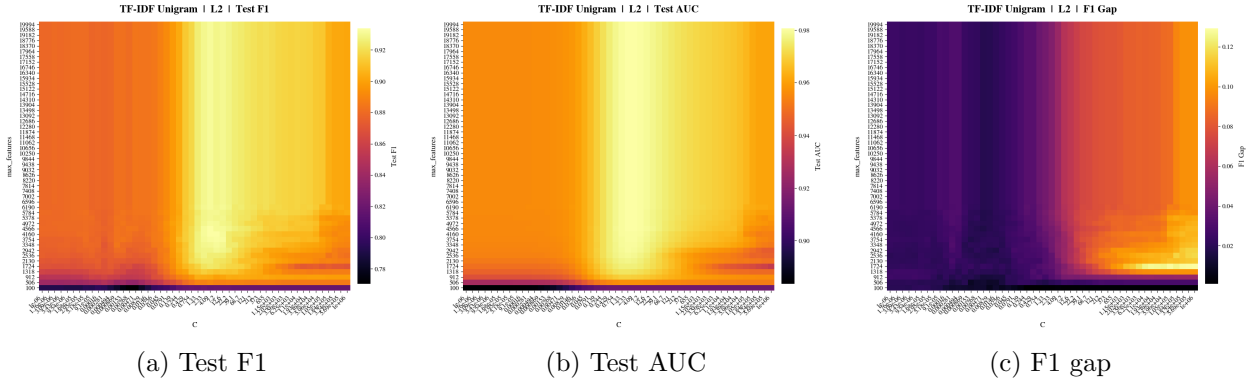


Figure 6: A representative example of the low-feature, weak-regularization corner. In the bottom-right region, predictive performance drops while the train–test gap rises, showing that severe feature truncation can become actively harmful once regularization is weak. Similar corner behavior appears in other heatmaps as well.

4.5 F1 and AUC Expose Different Kinds of Structure

Looking at F1 and AUC together changes how the heatmaps should be read. At first glance, the broad patterns look very similar in both metrics. The same large transition regions appear, and the same high-performing plateaus show up in many of the same places. That is important because it suggests that these broad features are real parts of the landscape rather than artifacts of one particular metric.

The more interesting difference appears in the smaller structures. In several BoW L2 heatmaps, there is a visible valley in F1 that is much weaker in AUC. This is useful because it shows that not every dark region in an F1 heatmap means the same thing. Some regions represent a genuine loss of performance in a broad sense, while others seem to reflect something more specific about how the model is making its final decisions.

To understand why this can happen, it helps to think about what logistic regression is actually producing. The model does not begin by directly saying “fake” or “real.” It first assigns each title a score, or equivalently a probability, based on the weighted sum of its features. Titles with higher scores are pushed more toward the fake class, and titles with lower scores are pushed more toward the real class. The final class label is assigned only after those scores are compared to a cutoff.

This is where the difference between F1 and AUC matters. AUC asks whether fake titles tend to receive higher scores than real titles overall. It is mainly about ranking. F1 asks whether the final class labels are correct after the scores are turned into yes-or-no predictions. It is therefore much more sensitive to what happens near the classification cutoff. A model can still rank the two classes reasonably well, which keeps AUC high, while at the same time producing less effective final classifications, which lowers F1.

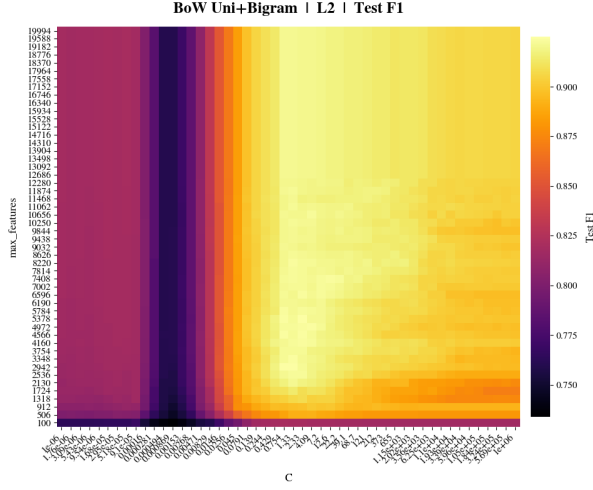
That seems to be what is happening in the BoW L2 valleys. The valley is strong in F1, but nonexistent in AUC. This suggests that the model has not completely lost the ability to keep fake and real titles apart. Instead, the scores appear to be less useful for making final decisions, even though the overall ordering of the two classes remains fairly good.

If many fake titles receive scores far above the cutoff and many real titles receive scores far below it, then the model will classify confidently and F1 will be strong. But if many titles from both classes are pushed closer to the cutoff, then small changes in score can flip the final class label. In that case, the final predictions become less reliable, even if the model is still ordering fake titles above real titles often enough to keep AUC relatively high.

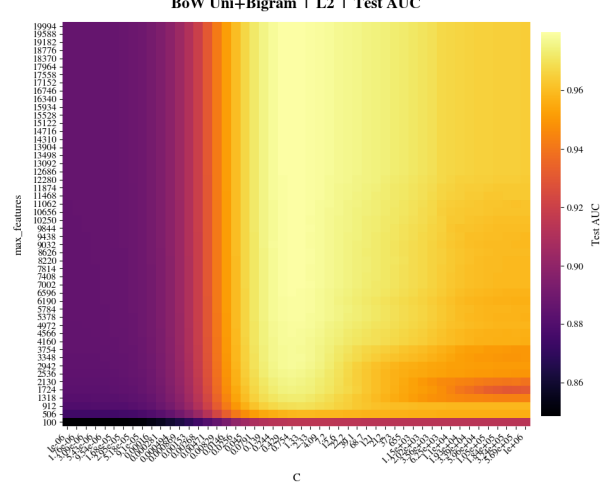
The fact that this valley appears in BoW but not TF-IDF suggests that the representation is part of the explanation. Under BoW, common words still enter the model through their raw counts. Under L2 regularization, many of these features can remain active with small but nonzero weights. That means the model may be using not only a few highly informative words, but also a wider background layer of weaker words that appear often across titles. Those weaker signals may still help preserve the overall ranking of the classes, which would keep AUC from falling too much. At the same time, they can make the final score distribution less decisive, so that more titles end up closer to the classification cutoff. That hurts F1 more than AUC.

TF-IDF changes this picture before logistic regression is even fit. Very common words are downweighted, while more distinctive words receive relatively more emphasis. As a result, the model relies less on broad background vocabulary and more on words that are actually useful for separating the classes. This likely reduces the same score-compression effect and helps explain why the valley structure is much weaker in the TF-IDF heatmaps.

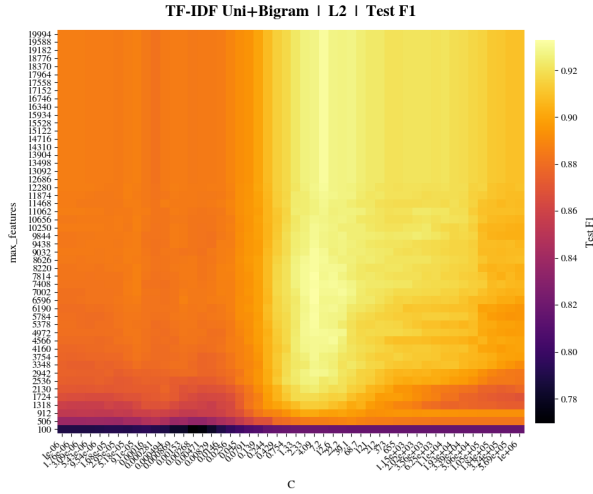
This comparison is one of the main reasons both metrics were needed. If only F1 had been used, it would have been easy to treat the valley as a full model failure. The AUC heatmaps show that the situation is more specific than that. The model still seems to preserve much of the class ordering in this region, but the final classifications become less effective. In that sense, the valley is not just a drop in performance. It is evidence that metric choice changes which parts of the model’s behavior become visible.



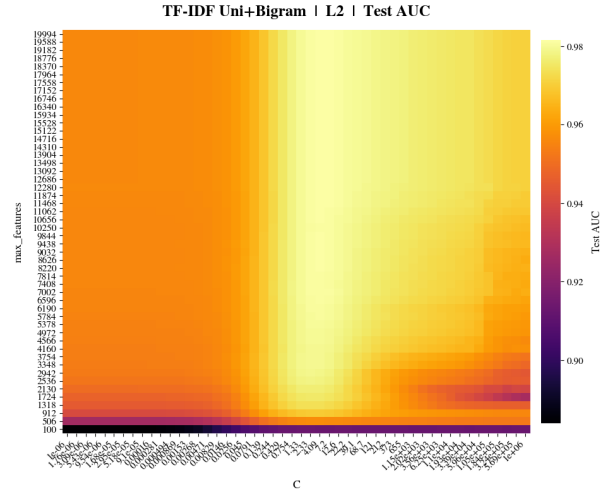
(a) BoW unigram + bigram, Test F1



(b) BoW unigram + bigram, Test AUC



(c) TF-IDF unigram + bigram, Test F1



(d) TF-IDF unigram + bigram, Test AUC

Figure 7: Comparison of F1 and AUC under BoW and TF-IDF with L2 regularization. The valley structure is most visible in the BoW F1 heatmap, disappears in BoW AUC, and is much less apparent in the corresponding TF-IDF heatmaps.

4.6 Generalization Gap Maps Identify Stable and Fragile Regions

The train–test F1 gap heatmaps add a third layer to the analysis. A small gap means that training and testing performance are close, while a larger gap means that the model fits the training data more strongly than the test data. By itself, however, a small gap is not always a good sign. It can reflect genuine stability, but it can also occur when the model performs poorly on both sets. For that reason, the gap maps have to be interpreted together with the F1 and AUC heatmaps.

This is especially clear in the L1 failure regions. On the left side of many L1 heatmaps, the gap is close to zero. At first glance, that might look stable. But in those same regions, both F1 and AUC are also poor, which means the model is failing on both training and test data. The gap is

small there not because the model has generalized well, but because it has learned very little on either set.

The more informative pattern appears as C increases. In many panels, F1 and AUC rise quickly and then level off, forming a broad high-performing plateau. The gap heatmaps show that this plateau is not uniform in stability. Even after predictive performance has mostly leveled off, the train-test gap continues to increase as regularization weakens. This means that the largest values of C are often not the most attractive practical settings, even when they still give strong raw performance.

Instead, the most useful region is often a moderate- C plateau: performance is already close to its best level, but the gap has not yet widened as much as it does in the weakly regularized extreme. This is one of the main advantages of looking at the gap heatmaps together with the performance heatmaps. The raw F1 and AUC maps show where the model performs well, while the gap maps show which parts of those high-performing regions are relatively stable and which are more fragile.

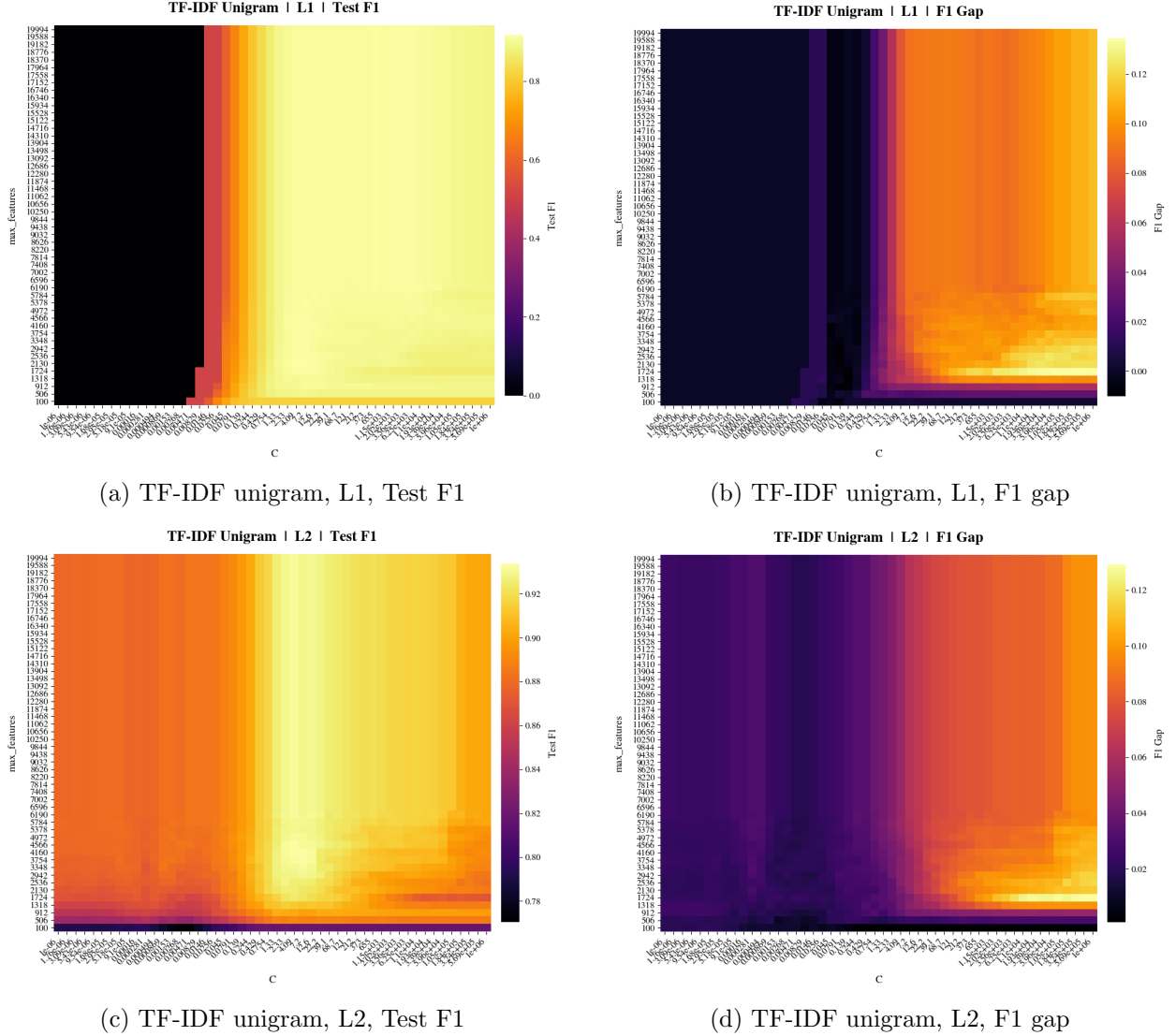


Figure 8: Comparing Test F1 and F1 gap shows that high performance does not automatically mean high stability. Under both penalties, strong performance emerges before the gap reaches its largest values, so the most useful region is often a moderate- C plateau rather than the weakest-regularization extreme.

4.7 Secondary Structures and Cautious Observations

In addition to the main collapse, plateau, and stability patterns, several smaller structures appear often enough to be worth noting. These are not emphasized as major conclusions, but they add detail to the overall picture of the landscape and help show that the heatmaps contain more than simple monotone trends.

One such pattern appears in several L1 gap heatmaps for unigram and unigram-plus-bigram models. To the left of the main high-performing region, there is often a broad field with very little activity, but inside that region a narrower vertical band appears in which the gap begins to increase.

This seems to mark a transition between total collapse and active fitting. In other words, the model is no longer completely failing, but it has not yet reached the more stable high-performing part of the landscape.

Another subtle difference appears in the shape of the L1 collapse boundary itself. In several BoW heatmaps, the boundary into the failure region looks vertical, while in the corresponding TF-IDF heatmaps it appears to taper slightly. This suggests that, under TF-IDF with L1 regularization, the onset of collapse may depend not only on the regularization strength but also somewhat on the available feature budget. A plausible explanation is that TF-IDF changes which features remain influential under L1 shrinkage, so increasing `max_features` can slightly shift the collapse boundary. At the same time, this observation should be treated cautiously. Since the C grid is discrete, an apparently sharper boundary in BoW may partly reflect limited resolution rather than a truly different underlying shape.

Taken together, these secondary structures do not overturn the main findings, but they show that the landscapes contain additional detail beyond the broad collapse, plateau, and stability patterns. Some regions appear to mark transitions between distinct fitting regimes, and some representation differences may affect the fine geometry of those transitions even when the broader performance story remains the same.

5 Discussion

The main value of this analysis is not that it identifies a single best logistic regression setting. The value of the heatmap approach is that it shows how model behavior changes across the parameter space and makes it possible to distinguish broad stable regions from narrow or fragile ones. That matters because the highest-scoring setting is not always the most useful one in practice.

Viewed this way, the results suggest that the most important choices in this setting are the feature representation and the regularization regime rather than the exact best hyperparameter pair. Unigram-inclusive models behave much more reliably than bigram-only models, which suggests that most of the useful signal in short news titles is already present at the single-word level. The heatmaps also show that regularization changes more than just the score itself. It changes the shape of the parameter space. Under L1, the model moves more abruptly from failure to usable performance, while under L2 the transition is smoother and the high-performing regions are broader.

The comparison between F1 and AUC adds another useful point. Some structures appear in both metrics, which makes them stronger evidence of a real pattern in model behavior. Others are much more metric-specific. In particular, the BoW L2 valleys appear clearly in F1 but disappear into a smooth gradient in AUC. That suggests that not every visible drop in F1 reflects a comparable loss in the model’s overall ability to keep the classes apart. This is a reminder that different metrics reveal different aspects of the same model.

The gap heatmaps also lead to a practical takeaway. High predictive performance by itself is not

enough to identify the most useful part of the parameter space. In many cases, strong performance appears before the train–test gap reaches its largest values. This suggests that the best practical region is often a moderate- C plateau rather than the weakest-regularization extreme. For this dataset, a unigram-inclusive representation with moderate regularization appears to offer the best overall tradeoff between predictive performance and stability.

At the same time, this approach has several limitations. First, the analysis is based on a single train–test split, so some smaller local patterns may partly reflect finite-sample variation. A stronger follow-up would be to repeat the experiment across multiple train–test splits and average the results. Second, the study focuses only on logistic regression. That was useful for understanding one model in depth, but it leaves open the question of whether the same landscape patterns would appear for other classifiers. Third, the heatmaps are shown with panel-specific color scales, which makes internal structure easier to see but can also exaggerate small numerical differences. In some cases, a color change may look more important than the underlying metric difference really is. A companion set of common-scale heatmaps or repeated-split summaries could help reduce that risk.

These limitations do not remove the main contribution of the project, but they do set the scope of the conclusions. The strongest findings are the broad recurring structures, not the smallest local anomalies. Even with those limits, the analysis shows that a landscape-based view can reveal aspects of model behavior that ordinary hyperparameter tuning would likely miss.

6 Conclusion

This analysis showed that the performance of logistic regression in title-based fake news classification is shaped by a few clear structural patterns rather than by isolated high-scoring settings. The heatmaps revealed a sharp low- C failure regime under L1, smoother transitions under L2, strong similarity between unigram and unigram-plus-bigram models, and weaker, less stable behavior for bigram-only representations. The most useful practical takeaway is that, in short text settings like this one, the best choice is not simply the weakest-regularization extreme or the most complex feature space. Instead, unigram-inclusive representations with moderate regularization and a feature budget of roughly 2,000 to 3,000 features, with around 2,000 as a particularly reasonable default, appear to offer the best overall tradeoff between predictive performance and stability. By contrast, bigram-only models generally require a larger feature range, roughly 5,000 to 6,000 features, yet still remain weaker overall. These ranges should be understood as approximate and specific to this dataset, but for datasets with similar short, lexical, title-like structure, they provide a useful starting point. More broadly, the project shows that mapping the full parameter space can reveal stable and unstable regions that ordinary hyperparameter tuning would miss.

References

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A Complete Heatmap Library

This appendix contains the full set of heatmaps used in the analysis. For each metric, heatmaps are grouped by feature representation. Within each figure, columns compare L1 and L2 regularization, while rows correspond to unigram, unigram-plus-bigram, and bigram-only feature spaces.

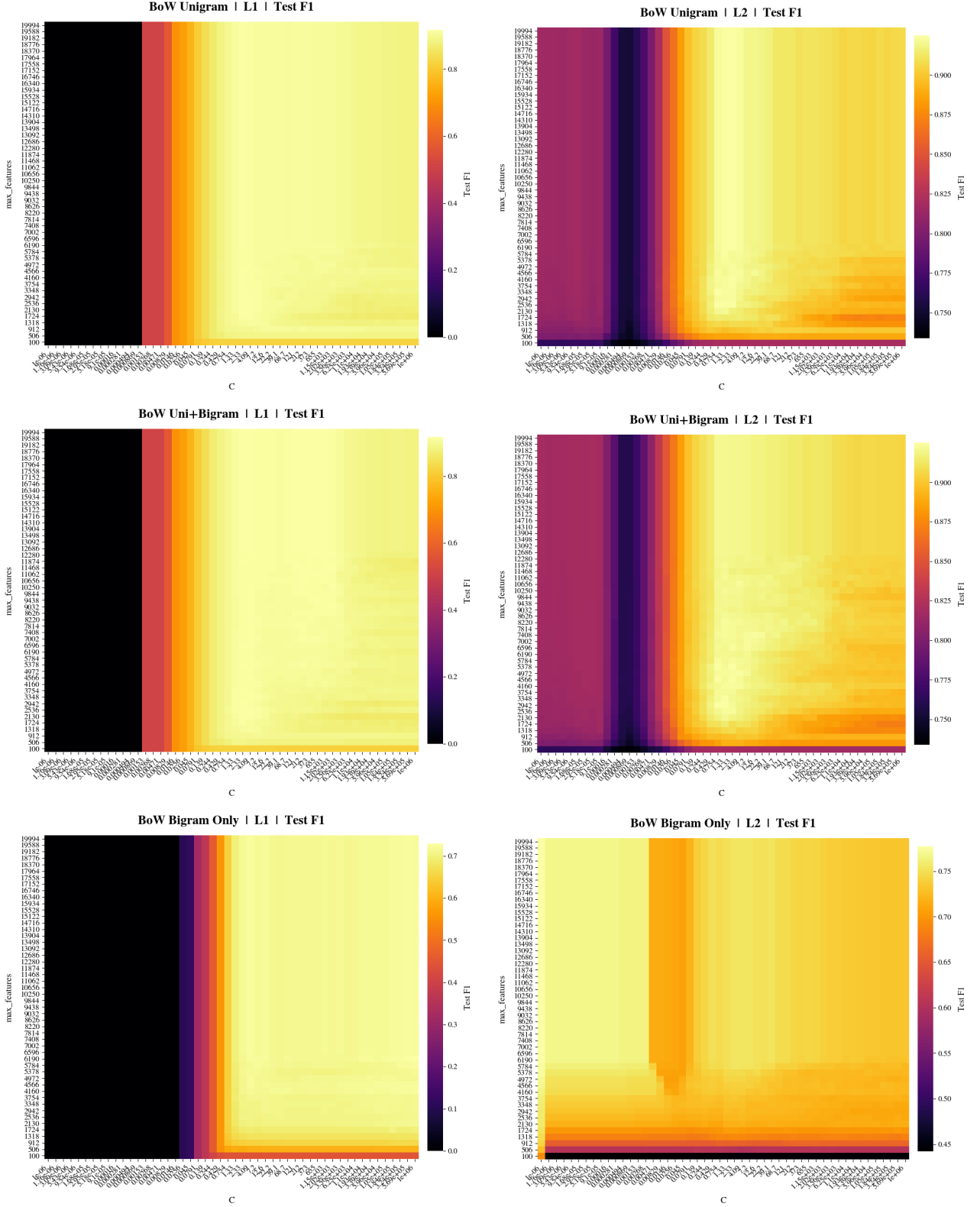


Figure 9: Complete Test F1 heatmaps for Bag-of-Words representations. Rows correspond to n-gram settings and columns compare L1 and L2 regularization.

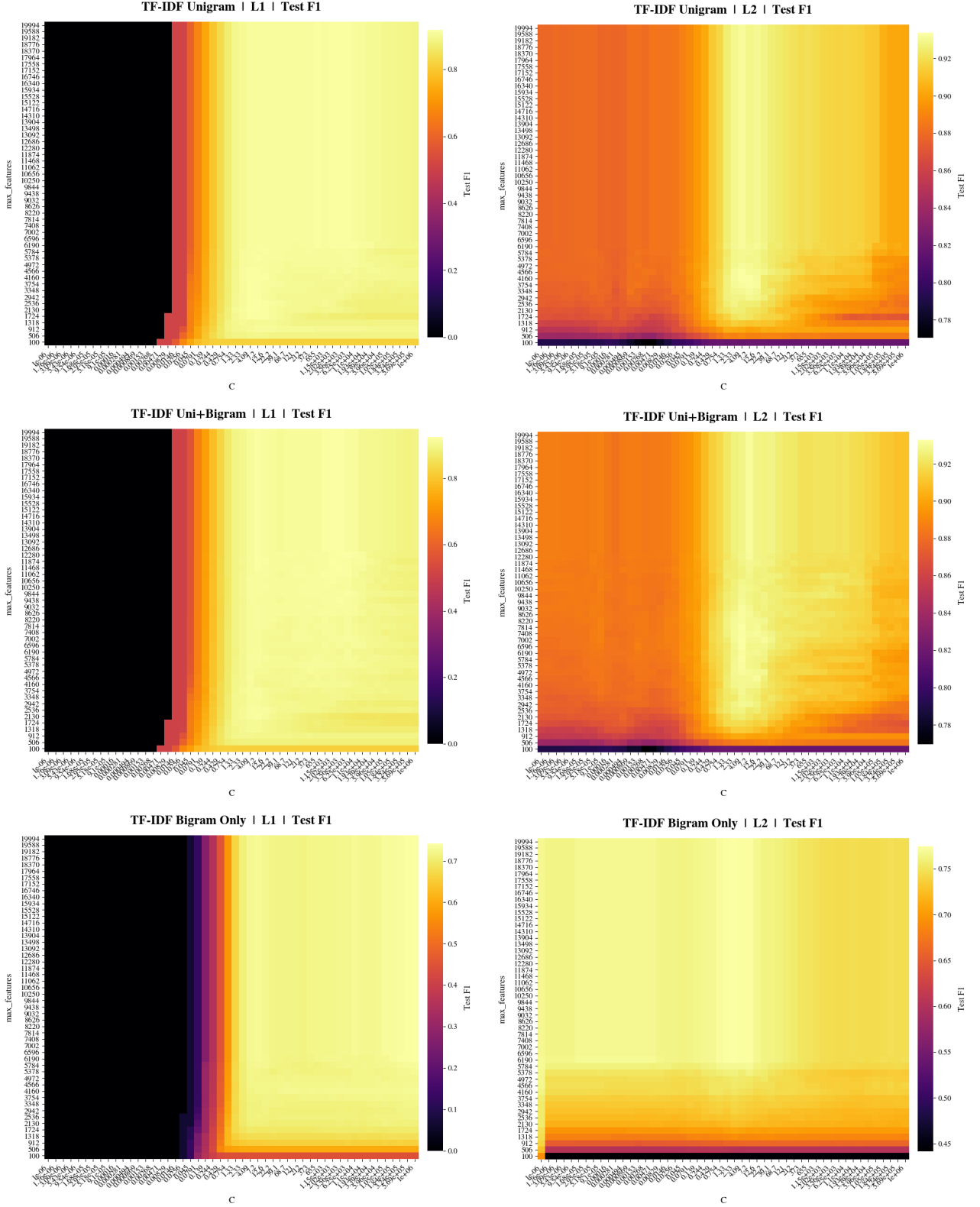


Figure 10: Complete Test F1 heatmaps for TF-IDF representations. Rows correspond to n-gram settings and columns compare L1 and L2 regularization.

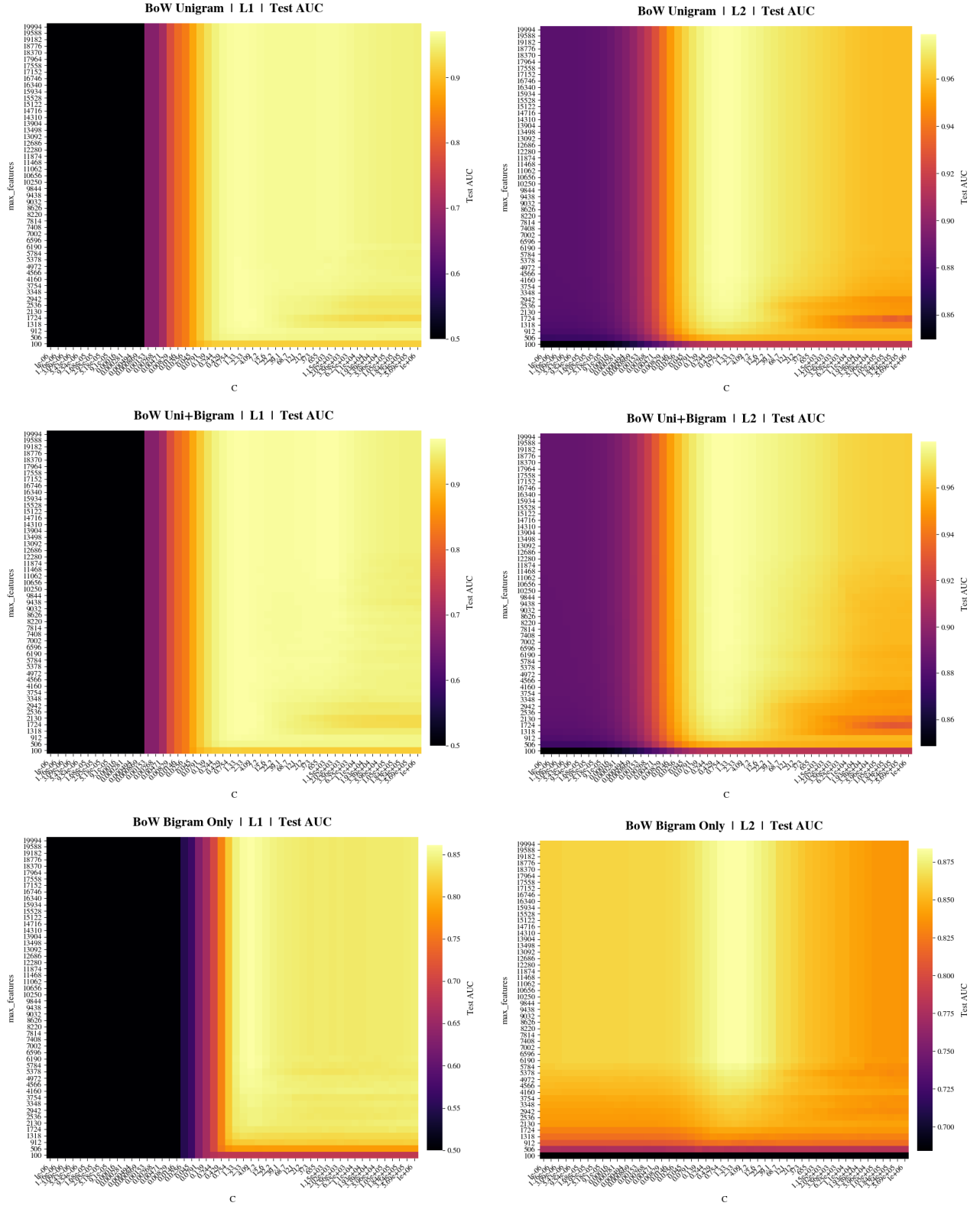


Figure 11: Complete Test AUC heatmaps for Bag-of-Words representations. Rows correspond to n-gram settings and columns compare L1 and L2 regularization.

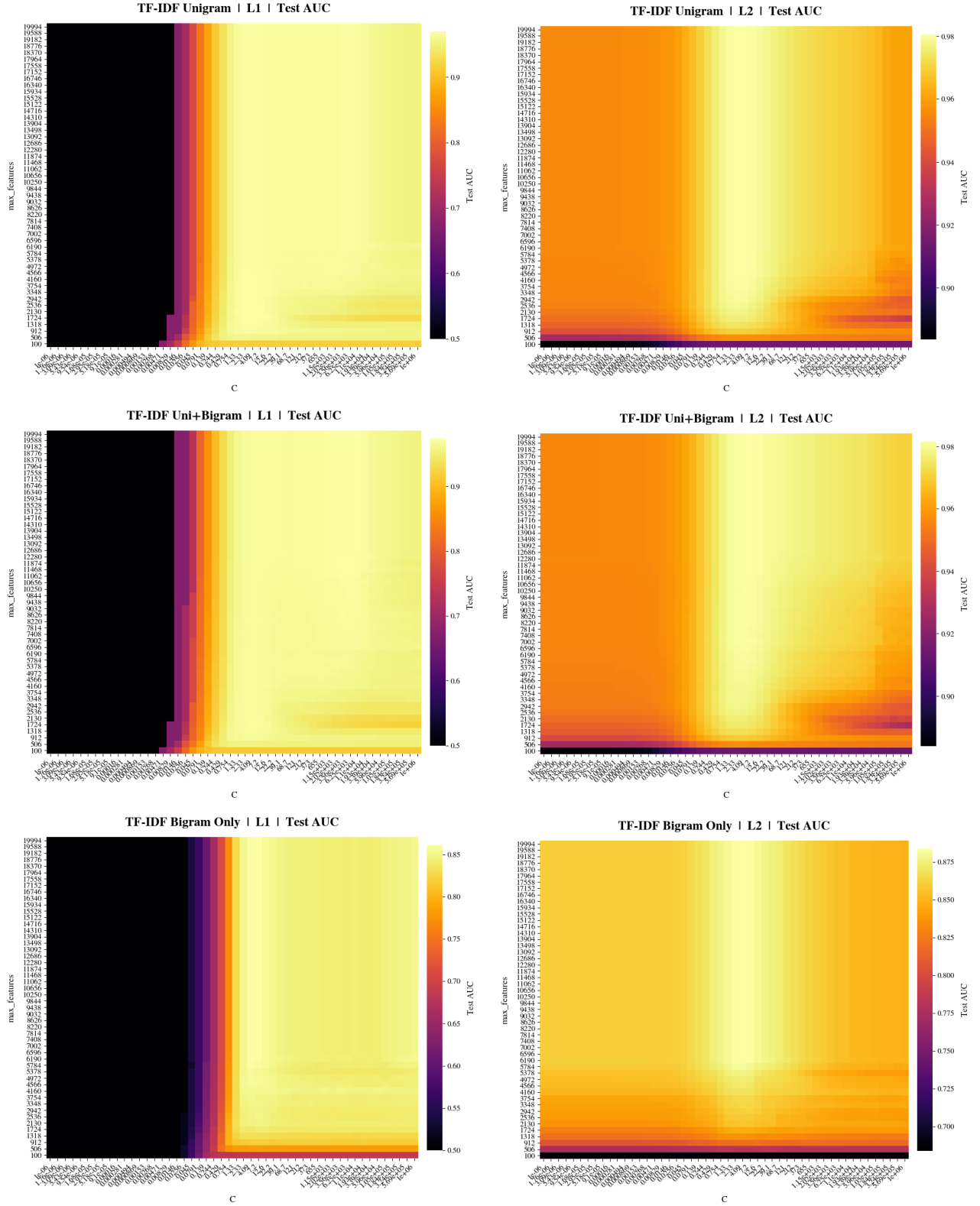


Figure 12: Complete Test AUC heatmaps for TF-IDF representations. Rows correspond to n-gram settings and columns compare L1 and L2 regularization.

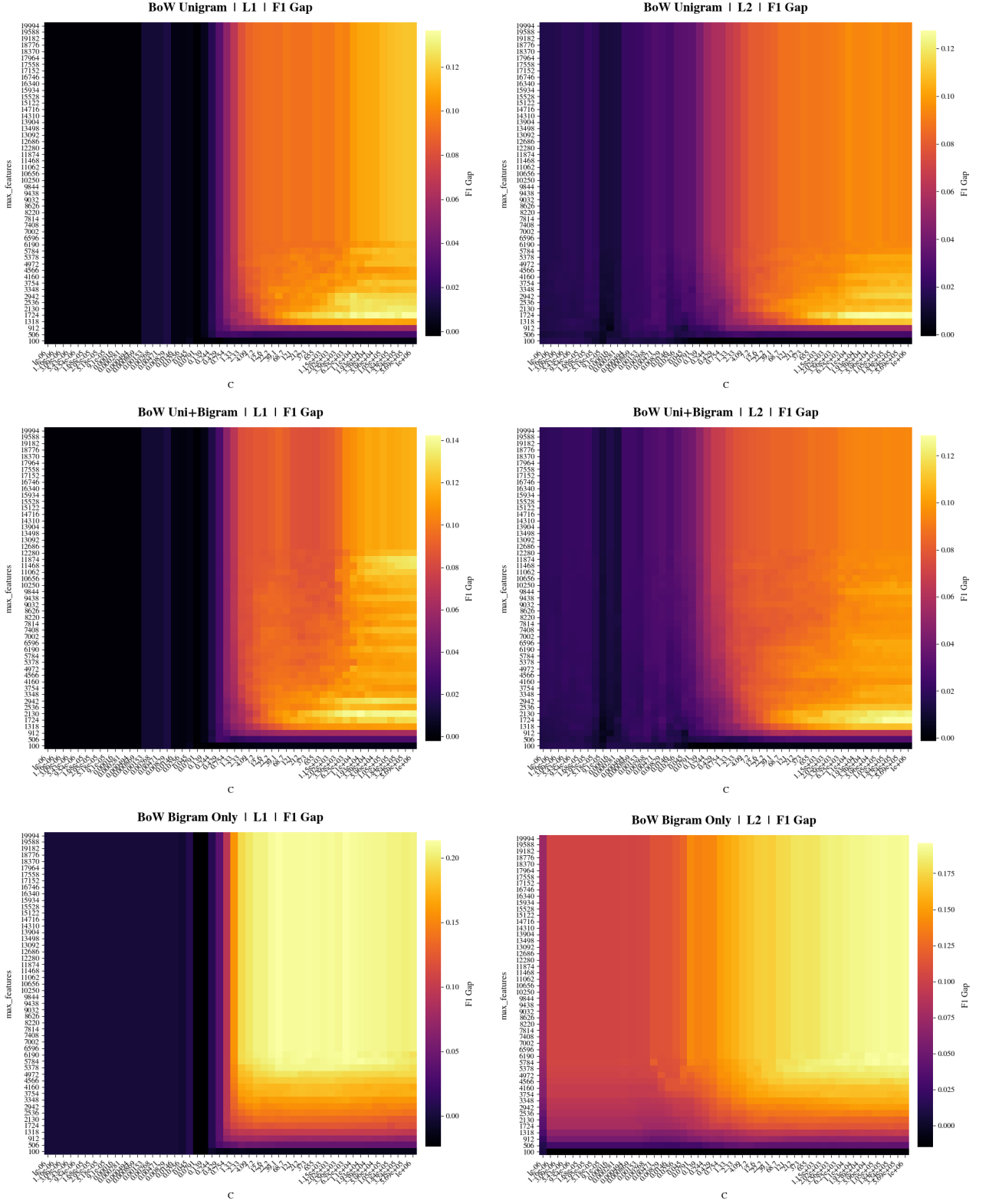


Figure 13: Complete F1 gap heatmaps for Bag-of-Words representations. Rows correspond to n-gram settings and columns compare L1 and L2 regularization.

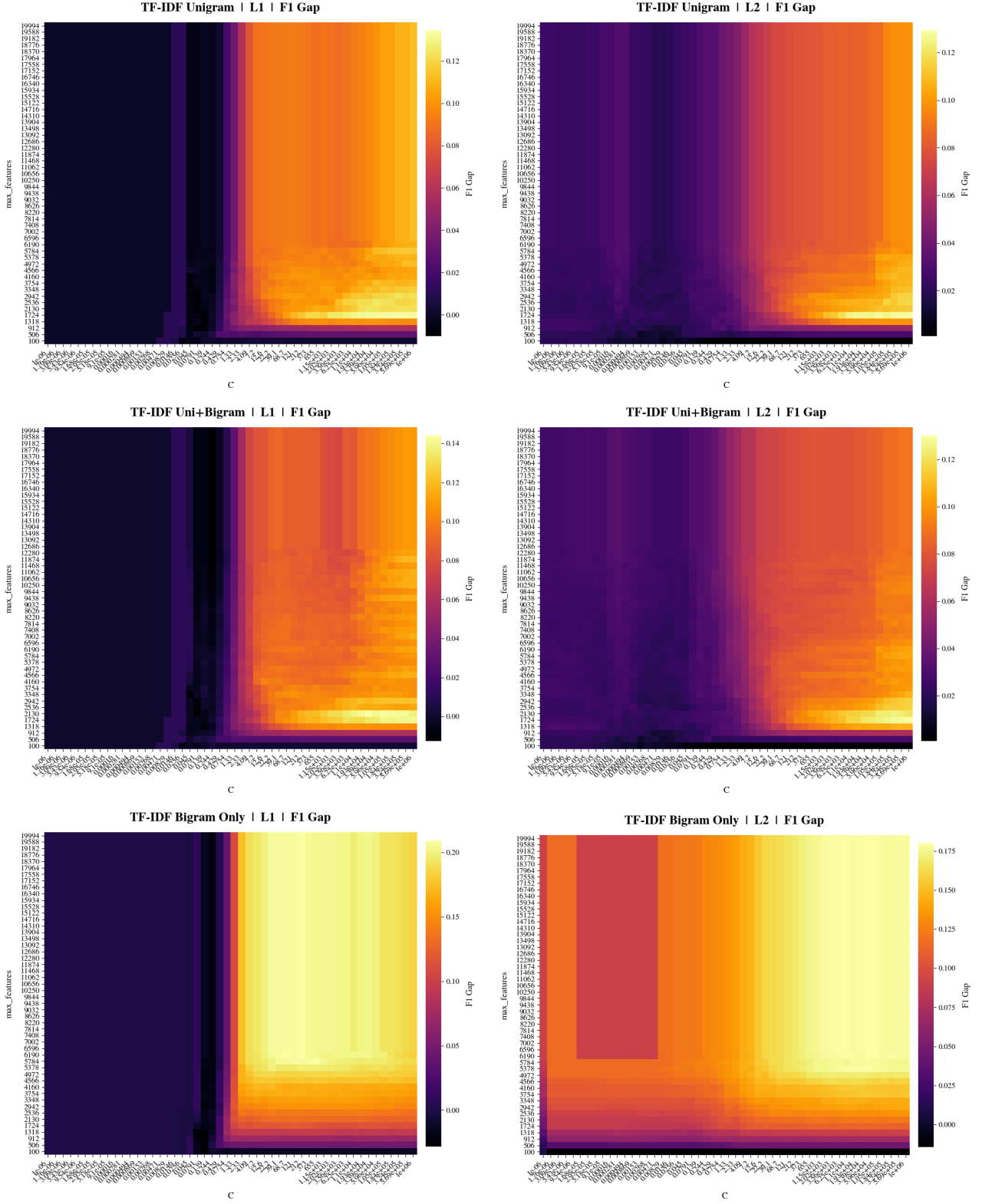


Figure 14: Complete F1 gap heatmaps for TF-IDF representations. Rows correspond to n-gram settings and columns compare L1 and L2 regularization.